

Feature Engineering Challenge

The business question

A shop has recorded one year of daily sales. Your task is to find what influences sales and when higher sales are expected.

What you are given

- A table of daily sales for 2025
- Weather information
- A coded date variable

What you must do

- Look for patterns and unusual values
- Suggest useful new variables
- Justify your reasoning from the data

Key idea

The useful variable may not be written directly in the dataset. You may need to create it from the columns you already have.

Why representation matters

Same data, different presentation

A pattern may be present in the data, but difficult to discover if the information is presented in an unhelpful form.

The agent trying to discover the regularity could be:

- a student inspecting the table by eye;
- a statistical model;
- a machine learning algorithm.

In all cases, the task becomes easier when the relevant structure is made more visible.

Feature engineering is the process of presenting the same underlying information in a more useful form.